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INTERPRETATIVE PERFORMANCE OF DIGITAL BREAST TOMOSYNTHESIS IN A TERTIARY HOSPITAL

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BACKGROUND

Screening mammography reduces breast cancer mortality [1]

Sensitivity reduced in dense breast tissue [2]

Combining Digital Breast Tomosynthesis (DBT) with full field digital mammography (FFDM) improves lesion visualization and cancer detection, especially in dense breasts [3]

OBJECTIVES

To evaluate DBT's **diagnostic performance** in detecting and characterizing breast lesions, using histopathology as the reference standard.

Secondary objectives:

- To determine the sensitivity, specificity and positive predictive value (PPV_3) of DBT in breast cancer.
- To measure PPV of BI-RADS 4 and 5 lesions on DBT
- To assess lesion conspicuity and margin details on DBT

METHODS

Single centre, retrospective cross-sectional study

Patient identification
n=1064 undergoing FFDM + DBT

Exclusion criteria
- HPE proven breast cancer
- Reconstructed breast/breast implants
- Previous breast surgery or mastectomy

Inclusion criteria
- Opportunistic screening
- Targeted screening (patients with risk factors)
- Diagnostic mammograms

Imaging & Lesion characterization
•ACR BI-RADS assessment (density mass, calcification
•Adjunct US for dense breast / suspicion

Reference standard
HPE via core biopsy or surgical specimen

Statistical analysis
Calculated sensitivity, specificity, PPV3 and BI-RADS performance using SPSS v28

RESULTS

Cohort, n= 1064

41.7% dense breast
BI-RADS C & D

Mass, n= 290
Calcifications, n=112
Asymmetries, n= 54

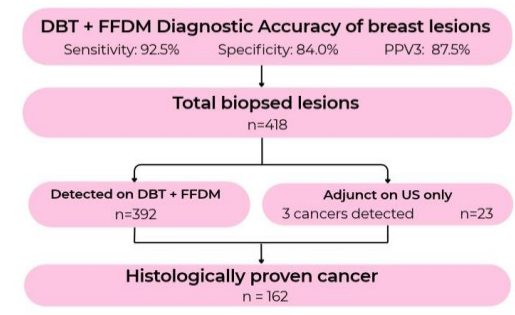

Figure 1: Diagnostic accuracy and lesion flowchart

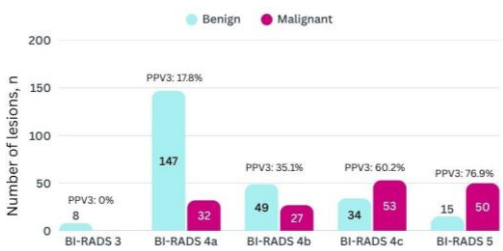

Figure 2: Bar chart demonstrating number of lesions according to BI-RADS category and its positive predictive value (PPV3) for each category

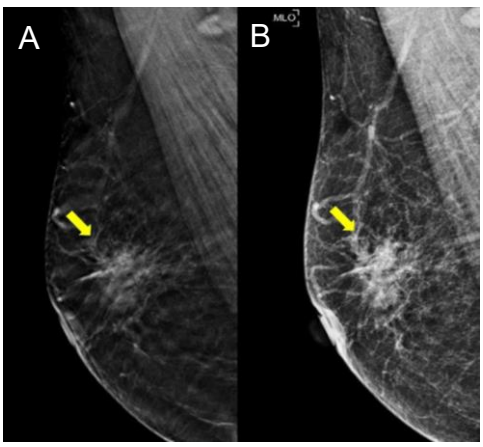

Figure 3: 58-year-old lady for screening. The DBT image in MLO (A) and the FFDM image in MLO (B) show asymmetrical density in the right upper region. The asymmetrical density is more pronounced in the tomosynthesis image. Histopathology of this lesion was revealed as DCIS.

CONCLUSION / TAKE-HOME MESSAGE

Combined DBT + FFDM demonstrates high diagnostic accuracy

- **Reduces false positive & unnecessary biopsies**
- **Overcomes 2D masking in dense breasts**
- **Optimizes diagnostic efficiency**

- **Integrate DBT as standard first line screening**
- **Conduct prospective longitudinal studies across regional Malaysian healthcare centers**

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